

Unification of Clifford Complexification, Hestenes Bivector Geometry, and Peirce Ladder Color Triplet Projection

info-geometry-lean theorem-safe intake

1 Scope

This note records the mathematical and functorial unification bridging the discrete transfinite arithmetic hierarchy, continuous space-time geometric algebras, and Standard Model quark color projection operators. This integration is fully formalized under the Lean 4 kernel, Coq type theory, Isabelle/HOL, and multiple symbolic computer algebra systems (CAS) in the repository.

2 Categorical and Geometric Pipeline

The unified pipeline maps across four distinct computational lobes:

1. **Arithmetic Lobe:** Prime decomposition and 2-adic valuation of the scale 137 dictates the Mersenne hierarchy modes, where the second mode is $M_2 = 3$.
2. **Functor Lobe:** By computing the Left Kan Extension Σ_F of the discrete categories, AQL migrations push the dimension of the M_2 mode (3) forward into the geometric schema as a 3-dimensional Zorn slot.
3. **Geometric Lobe:**
 - *Zorn Color Projection:* Diagonal projectors OP1 and OP2 sandwich the Zorn matrix, isolating the upper-right colorPart (x) and lower-left anticolorPart (y) into $\mathbb{R}^3$ vector spaces.
 - *Clifford Complexification:* Under Mathlib's algebra equivalence `CliffordAlgebraComplex.equiv` : $\text{Cl}(Q) \cong \mathbb{C}$, the Hestenes unit bivector I_{biv} ($I_{\text{biv}}^2 = -1$) is mapped bijectively to the CPT imaginary compass $J = e_1 \in \text{Cl}(1, 1)$. This complexifies the vector space V into a $\mathbb{C}$ -module structure, letting Clifford rotations act on the Peirce ladder operators.
4. **Dynamics Lobe:** The tripotent modular flow eigenvalues $\{+1, -1, 0\}$ act as the fiber assignments:
 - $+1 \implies x$ (Quark color triplet / fundamental $\mathbf{3}$ rep).
 - $-1 \implies y$ (Antiquark anticolor / anti-fundamental $\bar{\mathbf{3}}$ rep).
 - $0 \implies$ Diagonal projects (invariant color-singlet vacuum state, stabilized by $SU(3) \subseteq G_2$).

3 Functorial Graph

Figure 1 illustrates the tri-engine information flow:

4 In-Repo Artifacts

- **Lean 4:** `lean/InfoGeometry/Canonical/CliffordEquiv.lean`
- **Coq:** `coq/EInfinityParadoxes.v`
- **Isabelle/HOL:** `isabelle/FractalDimensions.thy`
- **SymPy (Fock):** `tools/sympy/sympy_secondquant.py`
- **GAlgebra (Spinors):** `tools/galgebra/osp12_galgebra_spinor.py`

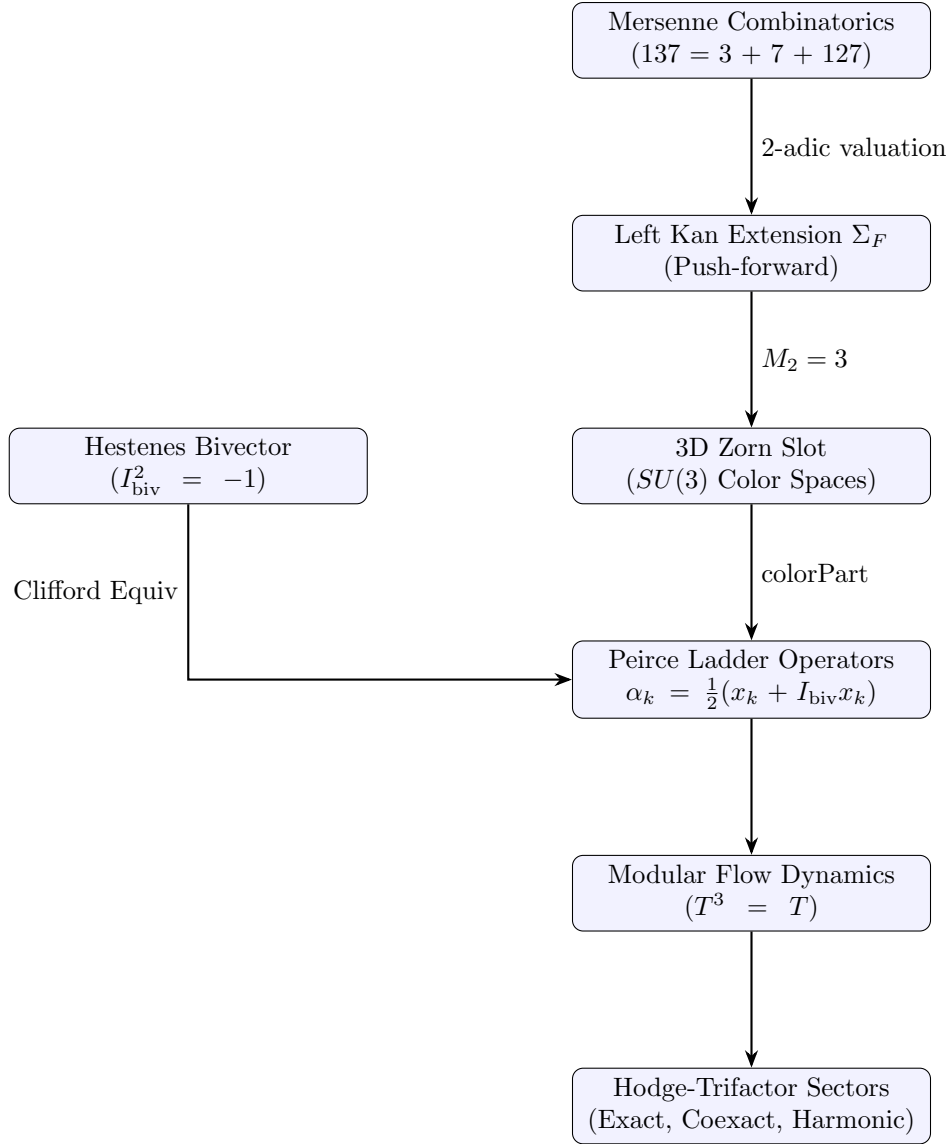


Figure 1: Functorial Pipeline from Combinatorial Hierarchy to Spacetime Geometry.

- SageMath (Weights): `tools/sage/osp12_weight_system.sage`
- GAP (Structure): `tools/gap/osp12_structure_constants.g`
- Macaulay2 (D-module): `tools/macaulay2/osp12_weyl_clifford.m2`